

Secure Enterprise Network Infrastructure Design and Implementation

BY : Mohamed Ashraf Mesbah

Introduction

This project is a complete enterprise network design and implementation created using a network simulation environment. The project focuses on building a secure and scalable network infrastructure using VLAN segmentation, routing, wireless connectivity, and firewall protection.

The network is designed to support multiple departments, servers, and wireless users while maintaining high performance, security, and manageability.

Project Objectives

- Design a scalable enterprise network.
 - Segment the network into multiple VLANs.
 - Provide both wired and wireless connectivity.
 - Secure the network using firewalls.
 - Implement routing and switching concepts.
 - Improve network performance and management.
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Project Components

1. Switches

The project uses multiple S3700 switches to connect end devices and distribute network traffic.

Functions of the Switches

- Connect PCs and servers.
- Support VLAN communication.
- Forward traffic efficiently.
- Reduce collisions and improve performance.

Number of switches used: 5

2. Firewalls

Two USG6000V firewalls are used to secure communication between different network zones.

Firewall Functions

- Filter network traffic.
 - Control access between VLANs.
 - Protect internal servers.
 - Separate trusted and untrusted networks.
 - Improve overall network security.
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3. Wireless Network

The wireless infrastructure includes:

- AP3030 Access Point
- AC6605 Wireless Controller

Purpose of the Wireless Network

- Provide Wi-Fi connectivity.
 - Support wireless clients.
 - Enable centralized wireless management.
 - Improve mobility inside the network.
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4. End Devices

The project includes:

- 5 PCs
- 3 Wireless STA devices

The devices are distributed across different VLANs for better organization and security.

5. Servers

Four servers are implemented in the network.

Server Name	Purpose
Guest Server	Provides guest services
Untrust Server	Simulates an untrusted network
Trust Server	Internal trusted server
Remote Server	Remote access services

Network Design

VLAN and Subnetting

The network is divided into multiple VLANs and subnets.

VLAN	Network Address	Purpose
VLAN 10	192.168.10.0 /24	User Department 1
VLAN 20	192.168.20.0 /24	User Department 2
VLAN 30	192.168.30.0 /24	Wireless Network
VLAN 40	192.168.40.0 /24	Servers
VLAN 50	192.168.50.0 /24	Additional Devices

Advantages of VLAN Segmentation

- Reduces broadcast traffic.
- Improves performance.
- Enhances security.
- Simplifies network management.

Sample IP Addressing

Device	IP Address
PC1	192.168.10.2
PC2	192.168.10.3
PC3	192.168.20.2
PC4	192.168.20.3
STA1	192.168.30.2
STA3	192.168.30.4
Guest Server	192.168.40.2
Trust Server	192.168.40.4
PC5	192.168.50.2

How the Project Works

1. End devices connect to switches.
2. Devices are assigned to different VLANs.
3. Routing is configured between VLANs.
4. Firewalls secure network communication.
5. Wireless users connect through the access point.
6. Users can access servers based on configured security policies.

Security Features

- Firewall protection between networks.
- Isolation of sensitive VLANs.
- Access control between departments.
- Guest network isolation.
- Ability to implement ACLs for traffic filtering.

Project Advantages

- Organized and professional network design.
 - Scalable infrastructure.
 - Support for wired and wireless communication.
 - Improved security and performance.
 - Easier troubleshooting and management.
 - Practical implementation of networking concepts.
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Possible Challenges

- Managing multiple VLANs.
 - Configuring inter- VLAN routing.
 - Securing wireless communication.
 - Managing firewall rules and policies.
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Technologies Used

- VLAN
 - Switching
 - Routing
 - Wireless Networking
 - Firewall Security
 - IP Addressing
 - Subnetting
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Conclusion

This project demonstrates a complete enterprise network infrastructure with secure communication, VLAN segmentation, wireless connectivity, and firewall protection. It applies both basic and advanced networking concepts, making it an excellent practical example for understanding modern network design and implementation.